

- 8 Dengue fever is caused by infection with the dengue virus. This virus is transmitted by mosquitoes.

The virus replicates inside the mosquito and is then passed onto a person in the saliva of the mosquito. The virus then replicates inside the infected person and causes the symptoms of the infection.

Symptoms include fever, headache and a rash. Complications include bleeding, low levels of platelets and low blood pressure.

It is estimated that globally, there are 390 million new cases of dengue infections per year. The infection rate is rising. This disease can be fatal.

Groups of scientists are working to develop vaccines, treatments and methods to reduce the spread of this virus.

- (a) The dengue virus is an RNA virus. The RNA is translated on the endoplasmic reticulum of the host cell into one polypeptide.

- (i) Describe how the viral RNA is translated.

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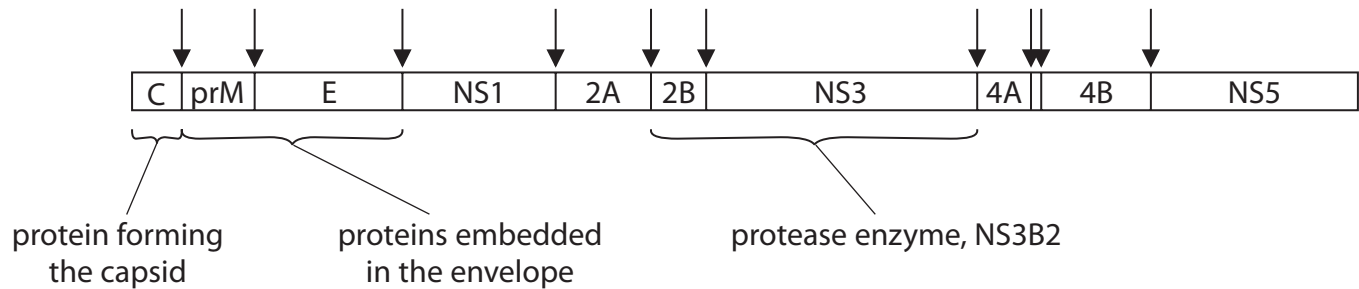
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- (ii) The diagram shows this polyprotein and the function of some of the individual proteins it contains.

The arrows indicate where the individual proteins need to be separated by the protease enzyme NS3B2.



One group of scientists has found that an inhibitor to this protease enzyme reduces the number of viruses produced in infected cells.

Suggest how this inhibitor could result in fewer viruses being produced.

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- (b) Another group of scientists showed that an extract from the leaves of the papaya plant helped in recovery from dengue virus infection.

They showed that the platelet and white blood cell counts were higher in people treated with the extract than in people given a placebo.

Explain how these changes may have helped in recovery from dengue virus infection.

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- (c) Another group of scientists infected the eggs of mosquitoes with *Wolbachia* bacteria.

The *Wolbachia* do not harm the mosquitoes but compete with the virus so that it is harder for it to replicate.

The bacteria can also be passed on to future generations of mosquitoes.

One study in Indonesia showed a 77% reduction in new cases of dengue infection after the introduction of *Wolbachia* into the mosquito population.

Describe **two** advantages of this method of disease control.

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(Total for Question 8 = 12 marks)

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